

2	(a)	Oscillating magnet causes a rate of change of magnetic flux linkage in the coil, by Faraday's Law, an EMF is induced. Since circuit is closed, by Lenz's Law, an induced current flows such that it produce a magnetic field to oppose the change in magnetic flux that causes it. <u>AND:</u> This magnetic field thus opposes the motion of the oscillating magnet. Since the motion is opposed, the amplitude is reduced. <u>Alternatively:</u> The induced current flows through the resistor causing a heating effect. This dissipate thermal energy which originated from the energy stored in the spring. Thus the motion of the magnet reduces in amplitude.	
	(b)	(i) $V_{rms} = V_0 / \sqrt{2} = 27.0 / \sqrt{2} = 19.092 \text{ mV}$ Therefore, $\langle P \rangle = 19.092^2 / 5$ $= 73 \text{ } \mu\text{W}$	
		(ii) $T = 2\pi / \omega = 2(3.14)/15.7 = 0.4 \text{ s}$ Labelled cosine square graph $P_o = V_o^2/R = (27 \times 10^{-3})^2 / 5 = 146 \text{ } \mu\text{W}$ OR $2\langle P \rangle = 146 \text{ } \mu\text{W}$	